AgriIntelligence: An Integrated Machine Learning-Based Agricultural Analytics and Decision-Support System for Crop Yield Prediction, Market Price Forecasting, and Weather Risk Assessment

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***Abstract*—Agriculture plays a vital role in food security and economic development, but agricultural planning is affected by changing weather conditions, fluctuations in market prices, and uncertainty in crop yield. This paper presents AgriIntelligence, an integrated agricultural analytics and decision-support system designed to support data-driven crop planning. The proposed system combines crop-yield, weather, and market-price datasets through data preprocessing, integration, exploratory data analysis, and feature engineering. Random Forest machine-learning models are used for crop-yield and market-price prediction. In addition, the system analyzes weather-related factors, including rainfall, temperature, and humidity, to assess potential agricultural risks. Based on the available yield, price, and weather information, a crop-recommendation module provides insights to support crop selection. An interactive dashboard developed using Streamlit brings together data visualizations, model predictions, weather-risk assessments, and crop recommendations in a single interface. The system aims to simplify access to agricultural insights and demonstrate how machine learning can support agricultural decision-making. Its current limitations include dataset coverage and dependence on the quality of available input data. Future enhancements may incorporate real-time weather forecasts, soil characteristics, satellite data, and comparisons with advanced machine-learning algorithms.**

***Index Terms*—agricultural analytics, crop yield prediction, decision support system, market price forecasting, random forest, Streamlit, weather risk assessment.**

I. Introduction

Agriculture plays a vital role in food security and economic development worldwide. However, agricultural decision-making is increasingly affected by uncertain crop yields, fluctuating market prices, and changing weather conditions. Accurate crop-yield prediction, market-price forecasting, and early identification of weather-related risks are important for helping farmers make informed decisions. Traditional agricultural planning often relies on historical experience and separate sources of information, which can make it difficult to evaluate multiple factors together. These challenges highlight the need for integrated, data-driven, and accessible agricultural decision-support systems.

This study presents AgriIntelligence, an integrated agricultural analytics and decision-support system. The key objectives are: [1] to preprocess, integrate, and analyze crop-yield, weather, and market-price datasets; [2] to develop Random Forest-based models for crop-yield prediction and market-price forecasting; [3] to assess weather-related agricultural risks using available environmental factors; and
[4] to provide crop recommendations and present analytical insights through an interactive Streamlit dashboard. The study also considers the limitations of the current system and discusses future improvements, including the use of real-world agricultural data, soil information, weather forecasting, and advanced machine-learning techniques.

[6] Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML), particularly algorithms such as Random Forest, have created opportunities [7]to analyze generate predictions[8] that assist in understanding crop yields and market-price trends. [10]Integrating these [9]Integrating these capabilities into a unified platform can improve the accessibility of agricultural insights and support more informed decision-making. [11]Integrating these informed decision-making.capabilities into a[12]Integrating these capabilities into a unified platform can improve the accessibility of [13]Integrating these capabilities into a unified platform can improve the accessibility of agricultural insights and support more informed decision-making.agricultural insights and support more informed decision-making. unified platform can improve the accessibility [15]of agricultural insights and support more informed decision-making.

II. Literature Survey

This work presents the application of machine learning techniques to agricultural analysis and crop-yield prediction. Machine-learning models can be trained using historical agricultural datasets containing crop-related and environmental information to estimate crop yields [1], [2]. Such predictions can help farmers and agricultural planners understand expected production and make informed decisions regarding crop selection and resource allocation. Ensemble methods such as Random Forest have also been applied successfully to yield estimation for individual crops such as sugarcane [3]. The integration of predictive analytics into agricultural systems can improve the accessibility of data-driven insights.

This approach also considers the integration of multiple agricultural data sources, including crop-yield records, weather information, and market prices. Combining these datasets provides a broader view of the factors affecting agricultural decisions. An interactive dashboard can present this information through visualizations, predictions, and analytical summaries, helping users interpret agricultural trends in a centralized interface.

Machine-learning methods can also be applied to market-price forecasting by analyzing historical price records and relevant agricultural features [4]. Understanding market-price trends can support crop-planning decisions and help users compare potential market opportunities. However, prediction quality depends on the availability, consistency, and representativeness of the underlying data.

Table I summarises the studies cited above and their relevance to the proposed system.

TABLE I
SUMMARY OF RELATED STUDIES

Ref.	Focus and technique	Relevance to AgriIntelligence
[1]	Systematic review of machine learning for crop-yield prediction	Reviews the features and algorithms used for yield prediction; motivates the yield module
[2]	Deep neural network for corn and soybean yield prediction	Shows that learning-based models can estimate yield from environmental data; deep models are a possible future comparison
[3]	Random Forest for sugarcane yield prediction	Supports the use of Random Forest for yield regression
[4]	Systematic review of machine learning for wheat, corn and rice price prediction	Informs the market-price forecasting module
This work	Integrated yield prediction, price forecasting, weather-risk assessment and crop recommendation	Combines the four tasks in one Streamlit dashboard using Random Forest models and rule-based components

Weather analysis is another important component of agricultural decision support. Factors such as rainfall, temperature, and humidity can influence crop growth and agricultural productivity. By examining these variables, an agricultural analytics system can identify potentially unfavorable conditions and communicate possible weather-related risks. Incorporating weather analysis alongside yield and market-price predictions provides a more integrated perspective on crop planning.

The AgriIntelligence project combines these concepts in a single agricultural analytics platform. Its main components include data preprocessing, data integration, exploratory data analysis, feature engineering, crop-yield prediction, market-price prediction, weather-risk assessment, and crop recommendation. The system presents the resulting insights through an interactive Streamlit dashboard. Further development could incorporate real-time agricultural information, soil characteristics, satellite observations, and additional predictive techniques.

III. METHODOLOGY

A. Feature Engineering

Feature engineering is an important stage in preparing agricultural data for machine-learning models. The purpose of this process is to identify and prepare relevant variables from crop-yield, market-price, and weather datasets for analysis and prediction. Depending on the available data, these variables may include crop type, historical yield, market price, rainfall, temperature, and humidity. The input features and target variable of each prediction model are listed in Table II.

TABLE II
INPUT FEATURES AND TARGET VARIABLES OF THE PREDICTION MODELS

Model	Input features	Target
Crop-yield model	State, district, crop, year, area, rainfall, temperature, humidity, market price	Crop yield
Market-price model	Crop, year, month, previous market price	Market price

Before feature engineering, the datasets undergo preprocessing to address issues such as missing values, duplicate records, inconsistent data types, and invalid entries. Data from different sources can then be integrated using appropriate common fields, where available. Exploratory data analysis is used to examine distributions, relationships, and trends in the data. These steps help prepare a consistent dataset for subsequent modeling.

B. Machine-Learning Model Development

Following data preparation and feature engineering, the selected features are used to train Random Forest models for crop-yield prediction and market-price prediction. Random Forest is an ensemble learning algorithm that combines predictions from multiple decision trees. It is suitable as an initial model for structured agricultural datasets because it can represent nonlinear relationships between input features and target variables. Random Forest has been reported to predict crop yield accurately in earlier studies.

The trained models can generate predictions from the input features supplied to them. Their performance should be assessed using appropriate evaluation metrics, such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2) for regression tasks. The metrics should be calculated using held-out test data and reported with the actual measured values; the measured results are given in Table VI.

C. Weather-Risk Assessment and Crop Recommendation

The weather-risk component analyzes available environmental variables, including rainfall, temperature, and humidity, to identify conditions that may be unfavorable for agricultural activities. In the current project, this component should be described as a rule-based or threshold-based assessment if it uses predefined conditions rather than a separately trained predictive model.

The crop-recommendation component combines relevant information from yield predictions, market prices, and weather-risk assessments to provide crop-selection insights. This integrated approach aims to help users compare agricultural options using several factors instead of relying on a single variable. The recommendations remain dependent on the quality and coverage of the available data and the rules or scoring method used by the system.

D. Model Evaluation

The crop-yield and market-price prediction models should be evaluated using appropriate regression metrics. MAE measures the average absolute difference between predicted and actual values, while RMSE gives greater weight to larger prediction errors. R^2 indicates how much of the variation in the target variable is explained by the model relative to a baseline.

For n test samples with actual values y_i, predicted values $\hat{y}_i$ and mean actual value $\bar{y}$, the metrics are computed as [6]:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \quad (1)$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \quad (2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2} \quad (3)$$

These metrics provide complementary information about prediction performance. Evaluation should use data that were not used to train the models. The resulting scores should be reported only after the models have been tested, so that the reported performance accurately represents the experiments conducted in the project.

E. Applications and Benefits

By bringing crop-yield prediction, market-price analysis, weather-risk assessment, and crop recommendation together, AgriIntelligence provides a centralized platform for agricultural decision support. The Streamlit dashboard allows users to explore available data, view analytical visualizations, review model predictions, and examine crop recommendations through an interactive interface.

Integrating these components can make agricultural information easier to access and interpret. The platform demonstrates how data analytics and machine learning can support agricultural planning by presenting multiple relevant factors in one place. It may also provide a useful foundation for future systems intended to assist farmers, researchers, and agricultural planners.

However, the current system should be considered a prototype rather than a fully validated agricultural advisory service, particularly if some datasets are synthetic or limited in geographic and historical coverage. Its predictions and recommendations should not be treated as guaranteed outcomes under real-world farming conditions.

IV. Data Resources

Reliable and diverse data are crucial for developing an effective machine-learning-based agricultural decision-support system. Agricultural productivity and crop-planning decisions are influenced by several factors, including crop yield, weather conditions, and market-price fluctuations. This research focuses on developing AgriIntelligence, an integrated agricultural analytics system that uses available datasets to support crop-yield prediction, market-price forecasting, weather-risk assessment, and crop recommendation.

A. Dataset Description

The datasets used in this project contain agricultural information related to crop yield, market prices, and weather conditions. These data provide the foundation for analyzing historical trends and developing predictive models. The integration of these different data sources enables the system to examine agricultural conditions from multiple perspectives. The datasets and their key attributes are summarised in Table III. The reliability and representativeness of the datasets are important considerations when interpreting the predictions and recommendations generated by the system.

Table III
Datasets Used in AgriIntelligence

Dataset	Key attributes	Used for
Crop-yield data	State, district, crop, year, production, area	Yield analytics and yield prediction
Weather data	Rainfall, temperature, humidity	Weather-risk assessment; yield-model inputs
Market-price data	Commodity price, market, year, month	Price analytics, price prediction and recommendation

A. Data Collection and Preprocessing

The AgriIntelligence system uses datasets containing information related to crop yield, weather conditions, and market prices. These datasets provide the foundation for understanding agricultural trends and developing predictive models. Before analysis, the data undergo preprocessing to improve consistency and suitability for machine learning.

B. Data Preprocessing

To improve data quality and prepare the information for analysis, the raw datasets undergo several preprocessing steps. These include handling missing values, identifying and removing duplicate records, converting data types into appropriate formats, and checking data consistency. Where appropriate, the datasets are integrated using common fields to create a unified dataset for further analysis. Exploratory data analysis and feature engineering are then performed to identify relevant variables and prepare the data for machine-learning models. The preprocessing operations are summarised in Table IV.

Table IV
Data Preprocessing Operations

Step	Operation and purpose
Missing values	Identify and handle missing entries so incomplete records do not affect the models
Duplicate records	Identify and remove repeated records
Data types	Convert fields to appropriate numeric, categorical and date formats
Consistency checks	Check the validity and consistency of values and remove invalid entries
Integration	Merge crop-yield, weather and market-price data using common fields
Exploratory analysis	Examine distributions, relationships and trends
Feature engineering	Select and prepare variables as machine-learning inputs

V. Proposed Framework: AgriIntelligence

AgriIntelligence is an integrated agricultural analytics and decision-support system designed to support crop planning through data analysis and machine learning. The system combines crop-yield, weather, and market-price datasets to generate useful agricultural insights. It applies data preprocessing and feature engineering techniques to prepare the datasets for analysis and uses Random Forest models for crop-yield prediction and market-price forecasting. In addition, the framework analyzes environmental factors, including rainfall, temperature, and humidity, to identify potentially unfavorable agricultural conditions.

The system integrates these analytical components into an interactive dashboard developed using Streamlit. Through the dashboard, users can explore agricultural data, examine predictions, review weather-risk assessments, and obtain crop recommendations based on the available information. The framework provides a unified approach to agricultural analysis and demonstrates how machine learning can support data-driven decision-making. The development methodology consists of several key stages.

The overall architecture of the system is shown in Fig. 1, and the structure of the prediction, risk-assessment and recommendation models is shown in Fig. 2. Table V summarises the techniques used by each module.

The preprocessing stage includes handling missing values, identifying duplicate records, converting data types, and checking data consistency. These operations help reduce data-quality issues that could affect subsequent analysis. Where appropriate, datasets are integrated using common fields to create a unified source for agricultural analytics.

The prepared data are then used for exploratory data

analysis and feature engineering. These processes help identify relevant variables, examine relationships among agricultural factors, and prepare suitable input features for model development.

TABLE V
SUMMARY OF SYSTEM MODULES

Module	Technique	Output
Preprocessing and integration	Cleaning, type conversion, merging on common fields	Unified dataset
Exploratory analysis and feature engineering	Trend and relationship analysis; feature preparation	ML-ready features
Crop-yield prediction	Random Forest regression	Predicted yield
Market-price prediction	Random Forest regression	Predicted price
Weather-risk assessment	Rule/threshold-based assessment of rainfall, temperature and humidity	Risk level (Low, Moderate, High)
Crop recommendation	Scoring rules combining yield, price and risk	Crop-selection suggestions
Dashboard	Streamlit	Interactive visualisations

B. Machine-Learning-Based Prediction

The proposed framework uses Random Forest algorithms to develop crop-yield and market-price prediction models. Random Forest is an ensemble learning technique that combines multiple decision trees to produce predictions. It is used in this project as an initial machine-learning approach for analyzing structured agricultural data.

The crop-yield prediction component estimates yield using the available agricultural features. The market-price prediction component analyzes relevant historical data to estimate market prices. These predictions provide information that can assist users in examining crop-production and market trends.

The performance of the models can be evaluated using regression metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). The measured results are reported in Section VI (Table VI).

C. Weather-Risk Assessment

Weather conditions can significantly influence agricultural productivity and crop-growing suitability. The AgriIntelligence framework analyzes available environmental variables, including rainfall, temperature, and humidity, to identify potentially unfavorable conditions.

The weather-risk component provides an additional source of information for agricultural planning. Its assessment is combined with the outputs of the prediction components to give users a broader understanding of the factors that may affect crop selection.

Where risk levels are determined using predefined thresholds or rules, the component functions as a rule-based assessment rather than a separately trained weather-prediction model. Its usefulness therefore depends on the suitability of the selected rules and the quality of the available weather data.

D. Crop Recommendation

The crop-recommendation component combines information from crop-yield predictions, market prices, and weather-risk assessments to provide crop-selection insights. By considering several factors together, the system aims to help users compare agricultural options more systematically.

The recommendations depend on the scoring rules, input features, and data available within the system. They should be interpreted as decision-support suggestions rather than guaranteed predictions of crop profitability or suitability.

E. Interactive Dashboard and Data Visualization

The AgriIntelligence dashboard is developed using Streamlit to provide an accessible interface for interacting with the system. It brings together agricultural data analysis, model predictions, weather-risk information, and crop recommendations.

The dashboard enables users to explore the available information without directly interacting with the underlying machine-learning code. Visualizations help present historical trends, while prediction and recommendation components make the analytical outputs easier to review.

The integrated dashboard serves as the user-facing component of the framework and demonstrates how agricultural data analytics can be organized into a single decision-support platform.

VI. RESULTS AND DISCUSSION

This section presents the AgriIntelligence dashboard and the evaluation of its prediction components. The project demonstrates the integration of agricultural data preprocessing, exploratory data analysis, machine-learning-based prediction, weather-risk assessment, and crop recommendation within a Streamlit dashboard. The framework provides a common interface for examining agricultural information and reviewing the outputs of its analytical components. Table VI reports the test-set performance of the Random Forest models, and Figs. 3–6 show the main dashboard pages.

TABLE VI
TEST-SET PERFORMANCE OF THE RANDOM FOREST MODELS

Model	MAE	RMSE	R^2
Crop-yield prediction	5867	6765	658756
Market-price prediction	7667	766	7676599

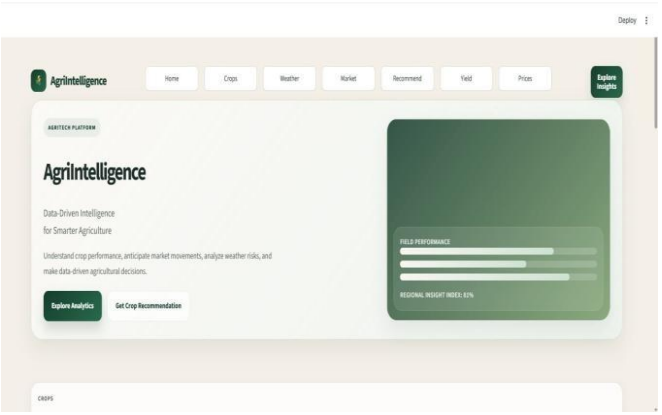


Fig. 3. AgriIntelligence dashboard home page.

Fig. 3 shows the landing page of the dashboard. A navigation bar gives access to the Home, Crops, Weather, Market, Recommend, Yield and Prices pages, and buttons on the page lead to the analytics and crop-recommendation views.

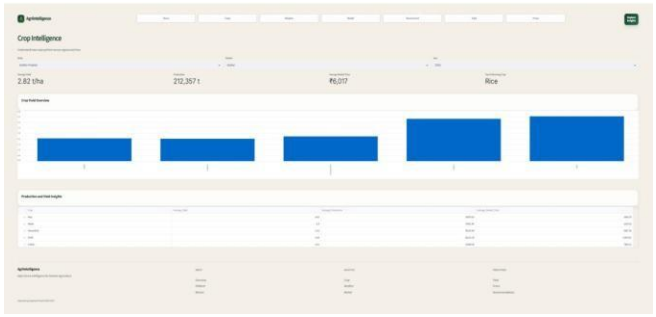


Fig. 4. Crop Intelligence page for agricultural data analysis.

Fig. 4 shows the Crop Intelligence page, where the crop data can be filtered by state, district and year. For the selected region it reports summary indicators (average yield, production, average market price and top-performing crop), a crop-yield bar chart, and a table of average yield, production and market price for each crop. The data behind this page pass through the preprocessing and validation steps described in Section IV.

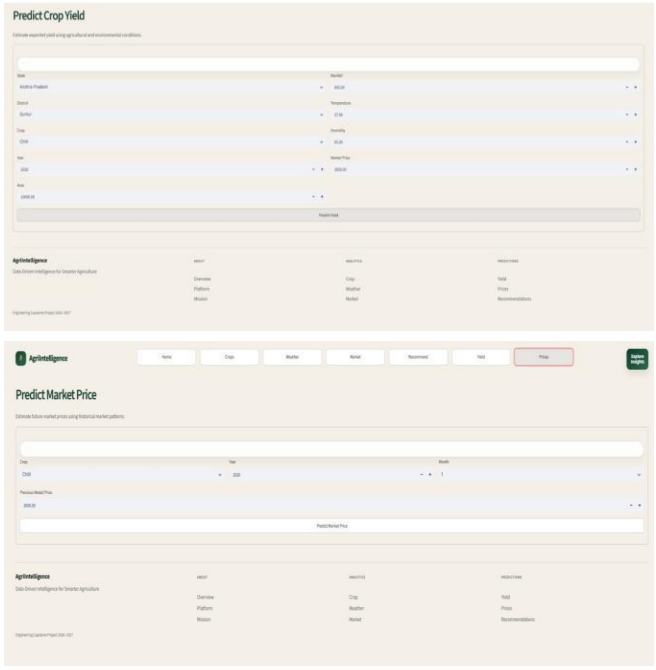


Fig. 5. Prediction interfaces: (a) crop-yield prediction (top) and (b) market-price prediction (bottom).

Fig. 5 shows the input forms of the two prediction pages. The yield page accepts state, district, crop, year, area, rainfall, temperature, humidity and market price, whereas the price page accepts crop, year, month and previous market price. The Random Forest models use these inputs to generate a prediction when the form is submitted. The reliability of the predictions should be judged from the test-set metrics in Table VI rather than from the presence of the interface alone.

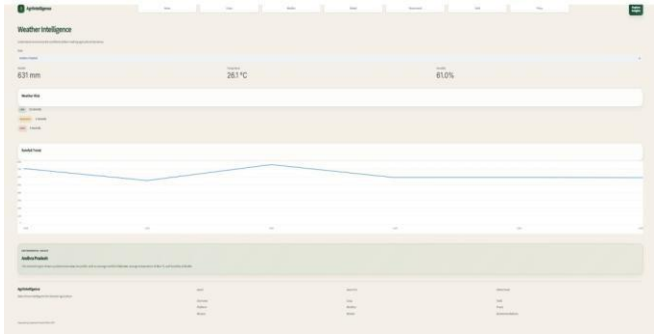


Fig. 6. Weather Intelligence page for weather-risk assessment.

Fig. 6 shows the Weather Intelligence page. For the selected state it displays average rainfall, temperature and humidity, the number of records in each risk category (Low, Moderate and High), a rainfall trend chart, and a short textual summary of the environmental conditions. These risk levels are combined with the yield and price outputs by the recommendation component.

Overall, AgriIntelligence demonstrates a unified approach to agricultural analytics by combining data preparation, predictive modeling, environmental assessment, and crop recommendation in one application. The integration of these components provides a foundation for exploring how data-driven methods can assist agricultural decision-making.

However, the practical reliability of the system depends on the quality and representativeness of the datasets, the predictive performance of the trained models, and the suitability of the recommendation rules.

VII. Conclusion and Future Scope

This paper presents AgriIntelligence, an integrated machine-learning-based agricultural analytics and decision-support system that combines crop-yield prediction, market-price forecasting, weather-risk assessment, and crop recommendation. By integrating crop-yield, weather, and market-price datasets, the proposed system provides a unified platform for analyzing agricultural information and supporting data-driven crop-planning decisions. Random Forest models are used for crop-yield and market-price prediction, while environmental factors such as rainfall, temperature, and humidity are analyzed to identify potentially unfavorable agricultural conditions. The system further combines these insights to generate crop recommendations and presents the results through an interactive Streamlit dashboard.

The integration of multiple agricultural analytics components into a single platform makes relevant information easier to access, visualize, and interpret. The dashboard enables users to explore available data, review predictions, assess weather-related risks, and examine crop recommendations within one application. This approach demonstrates the potential of combining machine learning and data analytics to support agricultural planning. However, the reliability and practical usefulness of the system depend on dataset quality, model performance, and the suitability of the recommendation methodology. If synthetic or demonstration datasets are used, further validation with representative real-world agricultural data is necessary before applying the system to actual farming decisions.

Future development should focus on improving prediction reliability, expanding data coverage, and increasing the transparency of the system's recommendations. Incorporating real-time weather forecasts, updated market prices, soil properties, soil moisture, and satellite-derived information could provide a more comprehensive understanding of agricultural conditions. Comparing Random Forest with other machine-learning algorithms, such as Gradient Boosting and XGBoost, could help identify models that perform better on the available datasets. In addition,

Further enhancements may include crop-profitability estimation, location-specific agricultural analysis, and automated alerts for significant changes in weather conditions or market prices. Developing a mobile-friendly interface could also improve accessibility for users in different agricultural settings. Finally, testing the system with larger and more diverse real-world datasets, evaluating its performance across different regions and seasons, and gathering feedback from intended users would help assess its practical value. These improvements could support the evolution of AgriIntelligence into a more reliable, transparent, and comprehensive agricultural decision-support platform.

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